

Computer Organization and Architecture

Chapter 1 Introduction



Computer Organization and Architecture

Computer Organization - the internal arrangements of a computer, which includes the design of the processor, memory and input/output circuits.

Computer Architecture describes features of a computer family (notably the instructions) and not the specific implementation, just like architecture of a house might be described as Victorian.

Some do not differentiate between ***computer organization*** and ***computer architecture***.

Architecture & Organization 1

- **Architecture** is those attributes visible to the programmer
 - Instruction set, number of bits used for data representation, I/O mechanisms, addressing techniques.
 - e.g. Is there a multiply instruction?
- **Organization** is how features are implemented. refers to the operational units and their interconnections that realize the architectural specifications.
 - Control signals, interfaces, memory technology.
 - e.g. Is there a hardware multiply unit or is it done by repeated addition?

Architecture & Organization 2

- All Intel x86 family share the same basic architecture
- The IBM System/370 family share the same basic architecture
- This gives code compatibility
 - At least backwards
- **Organization** differs between different versions

Structure & Function

- **Structure** is the way in which components relate to each other
- **Function** is the operation of individual components as part of the structure

Function

- All computer functions are:
 - Data processing
 - Data storage
 - Data movement
 - Control

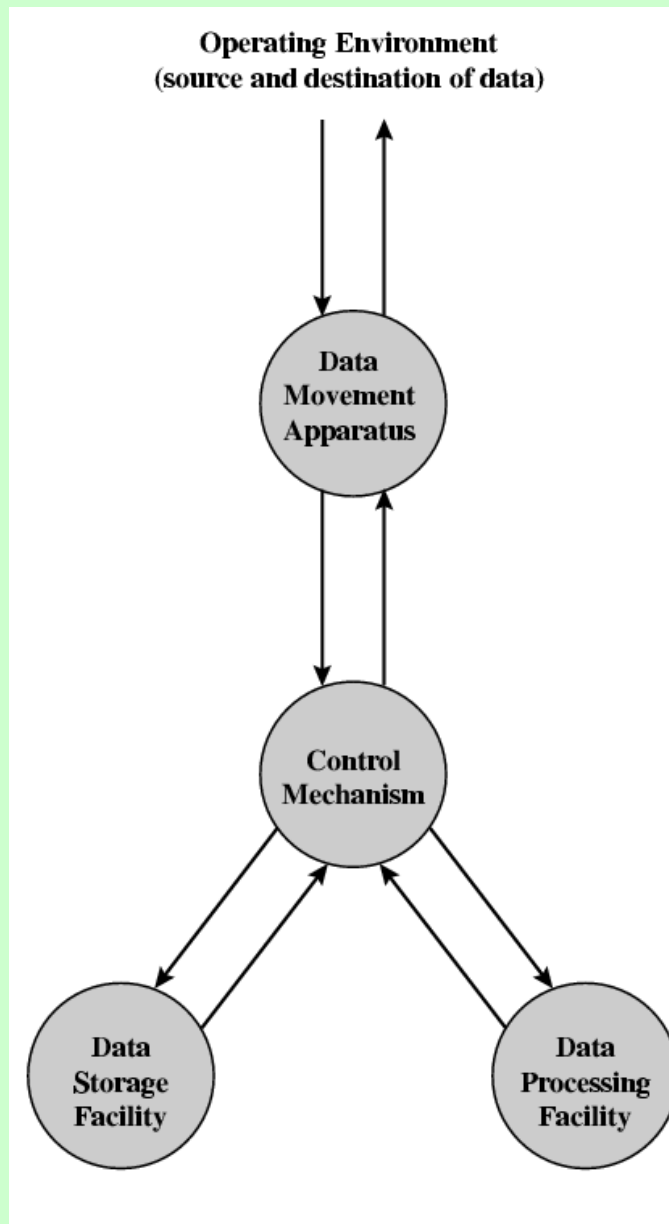
Function

- There are four basic functions that a computer can perform:
 - Data processing
 - Data may take a wide variety of forms and the range of processing requirements is broad
 - Data storage
 - Short-term
 - Long-term

Function

- Data movement
 - Input-output (I/O) - when data are received from or delivered to a device (peripheral) that is directly connected to the computer
 - Data communications – when data are moved over longer distances, to or from a remote device
- Control
 - A control unit manages the computer's resources and orchestrates the performance of its functional parts in response to instructions

Functional View



Functional View

The computer must be able to **move data** between itself and the outside world.

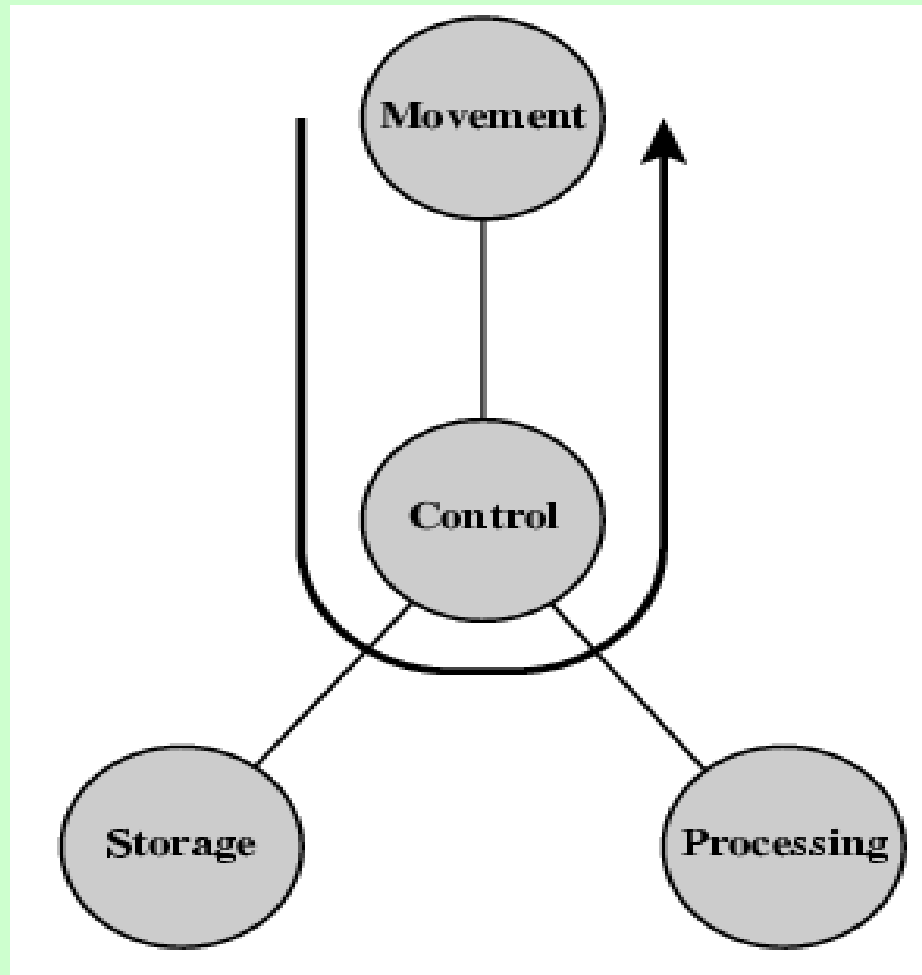
The computer's operating environment consists of devices that serve as either sources or destinations of data.

When data are received from or delivered to a device that is **directly** connected to the computer, the process is known as ***input–output (I/O)***, and the **device** is referred to as a ***peripheral***.

When data are moved over longer distances, to or from a remote device, the process is known as ***data communications***.

Operations (a) Data movement

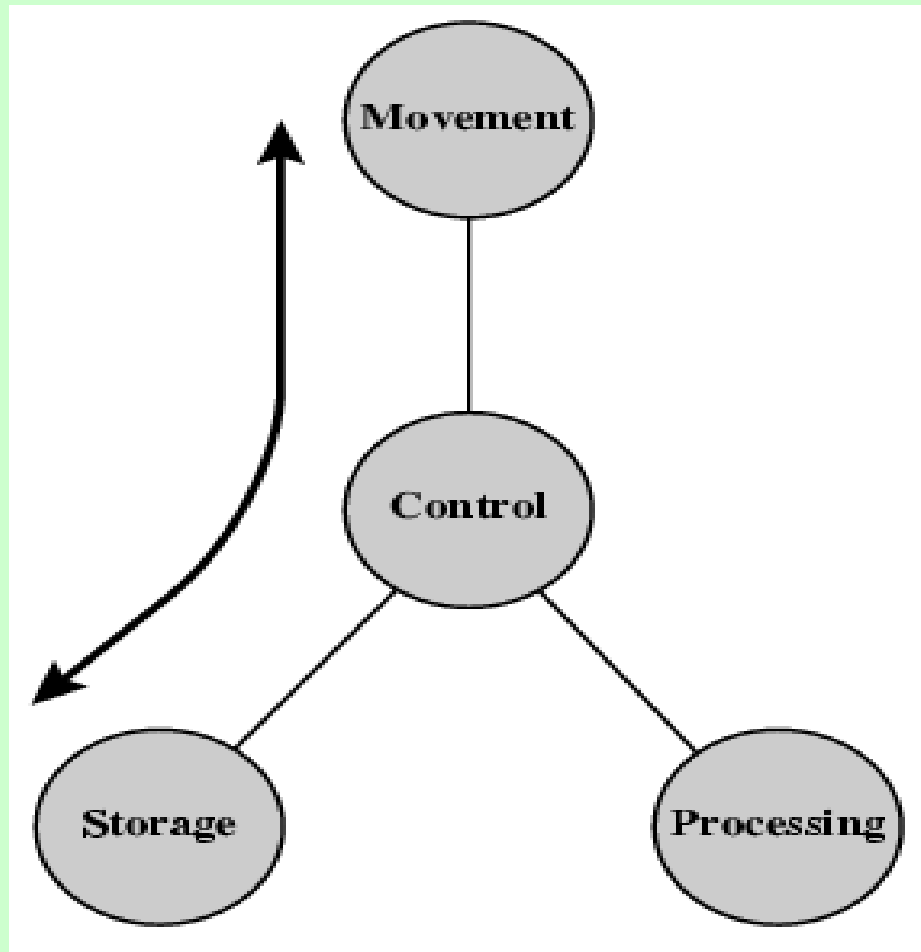
Transferring data from one peripheral or communications line to another
e.g. keyboard to screen



Operations (b) Storage

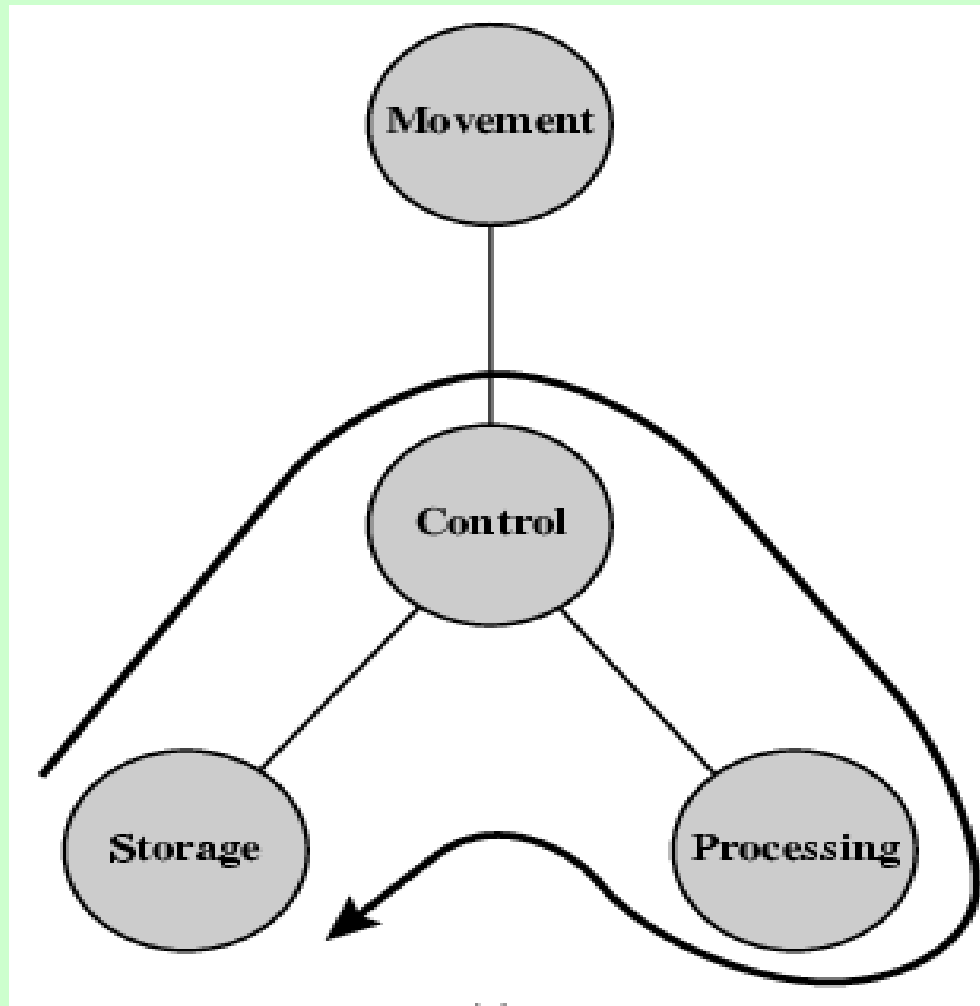
Data transferred from the external environment to computer storage (read) and vice versa (write)

e.g. Internet download to disk



Operation (c) Processing from/to storage

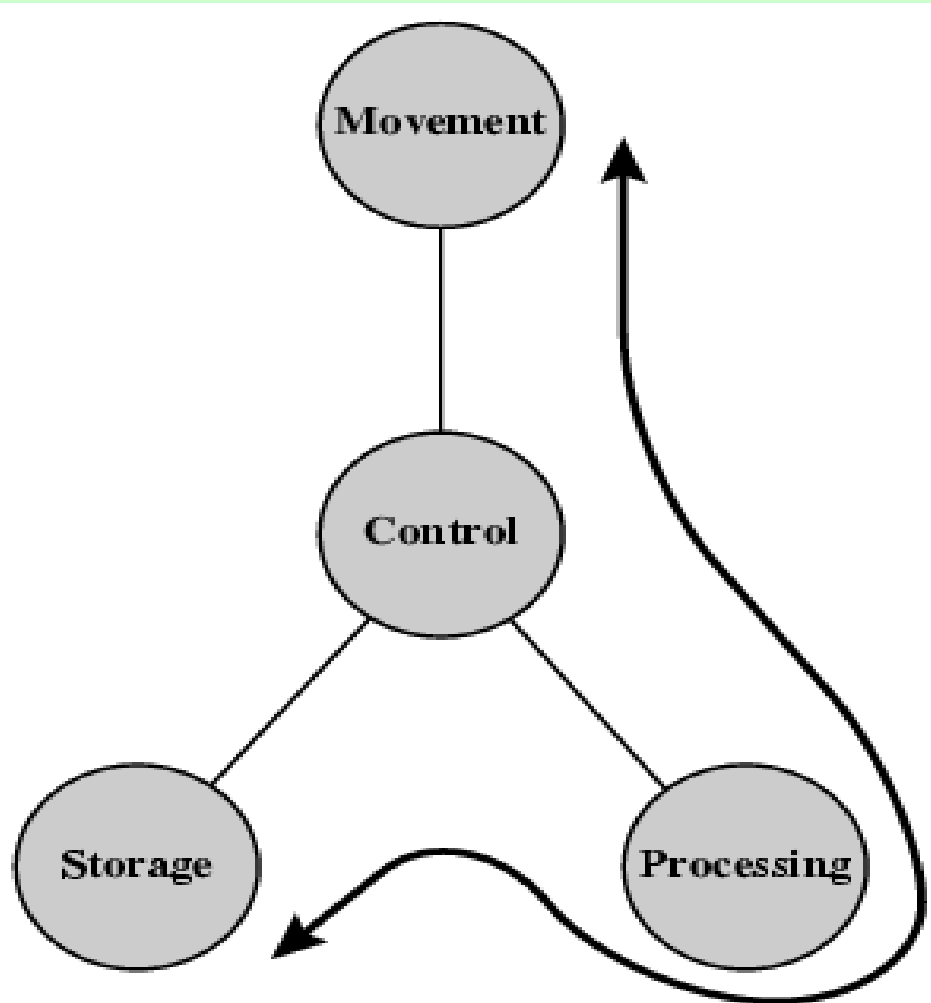
Show operations involving data processing
e.g. updating bank statement



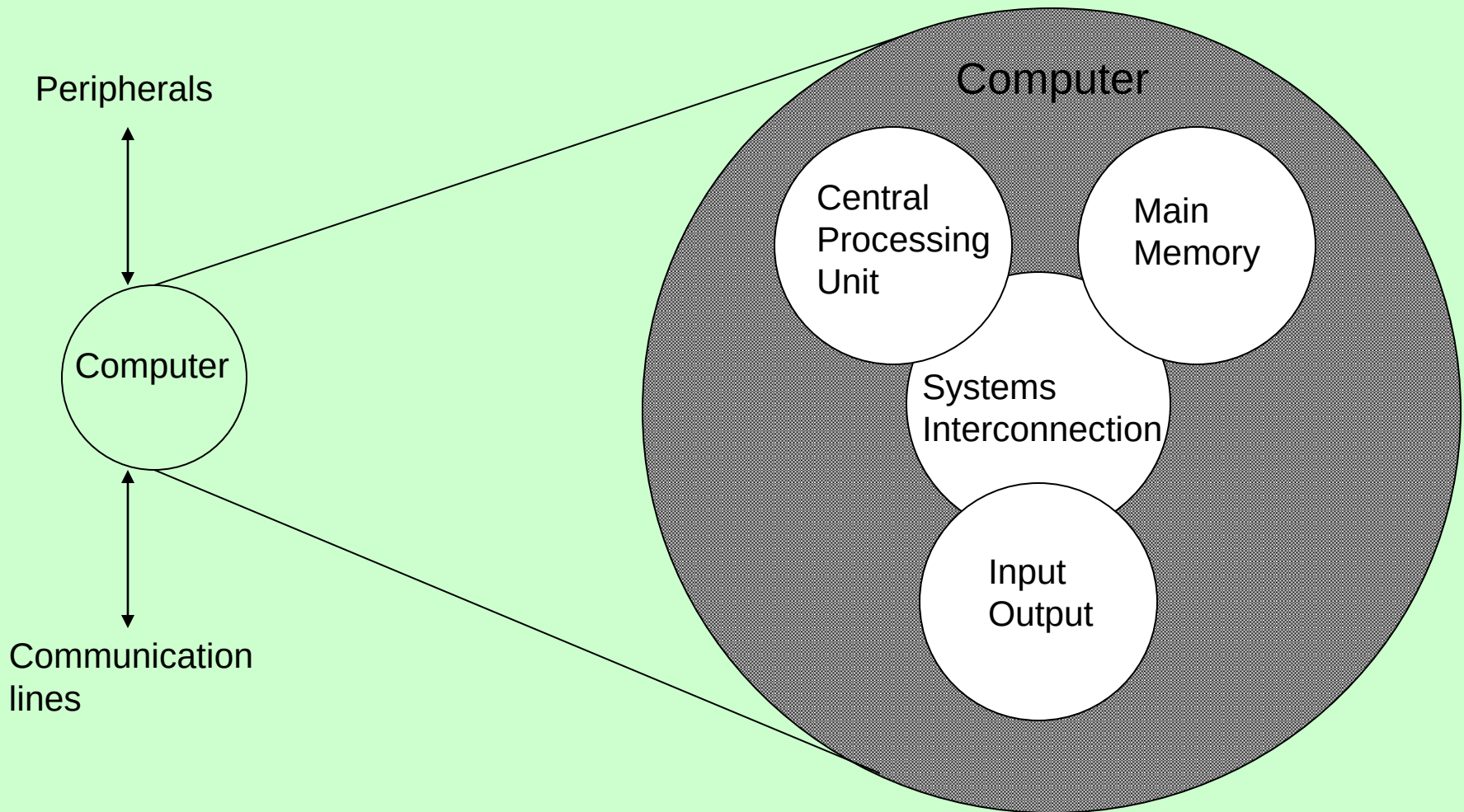
Operation (d)

Processing from storage to I/O

Show operations involving data processing
e.g. printing a bank statement



Structure - Top Level



The Four Main Structural Components of the Computer:

- C P U – controls the operation of the computer and performs its data processing functions
- Main Memory – stores data
- I/O – moves data between the computer and its external environment
- System Interconnection – some mechanism that provides for communication among C P U, main memory, and I/O

CPU Major structural components:

Control Unit

Controls the operation of the C P U and hence the computer

- **A**rithmetic and **L**ogic **U**nit (A L U)

Performs the computer's data processing function

- Registers

Provide storage internal to the C P U

- C P U Interconnection

Some mechanism that provides for communication among the control unit, A L U, and registers

Multicore Computer Structure (1 of 2)

Central processing unit (C P U)

- Portion of the computer that fetches and executes instructions
- Consists of an A L U, a control unit, and registers
- Referred to as a processor in a system with a single processing unit

Core

- An individual processing unit on a processor chip
- May be equivalent in functionality to a C P U on a single-C P U system
- Specialized processing units are also referred to as cores

Multicore Computer Structure (2 of 2)

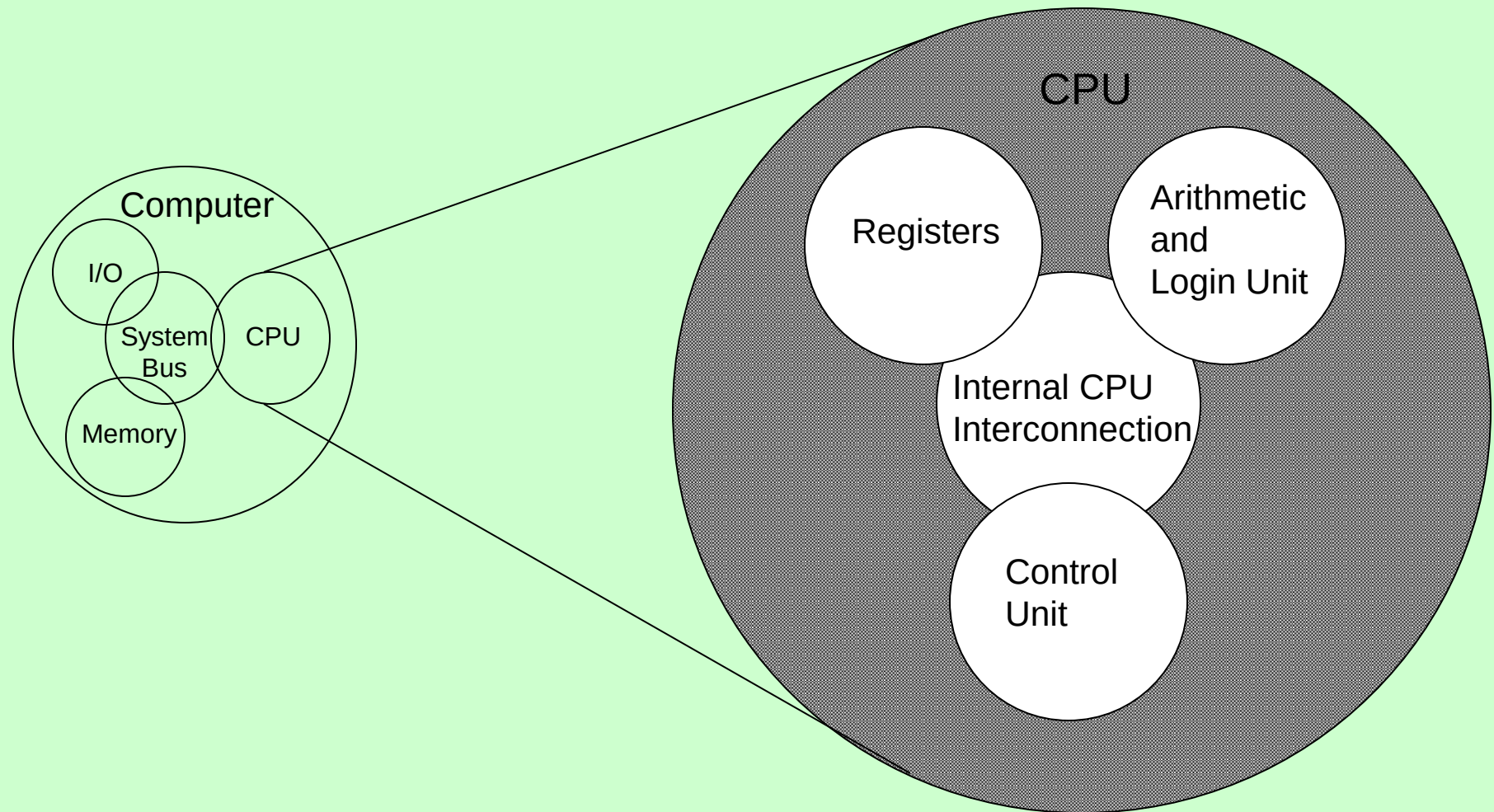
Processor

A physical piece of silicon containing one or more cores

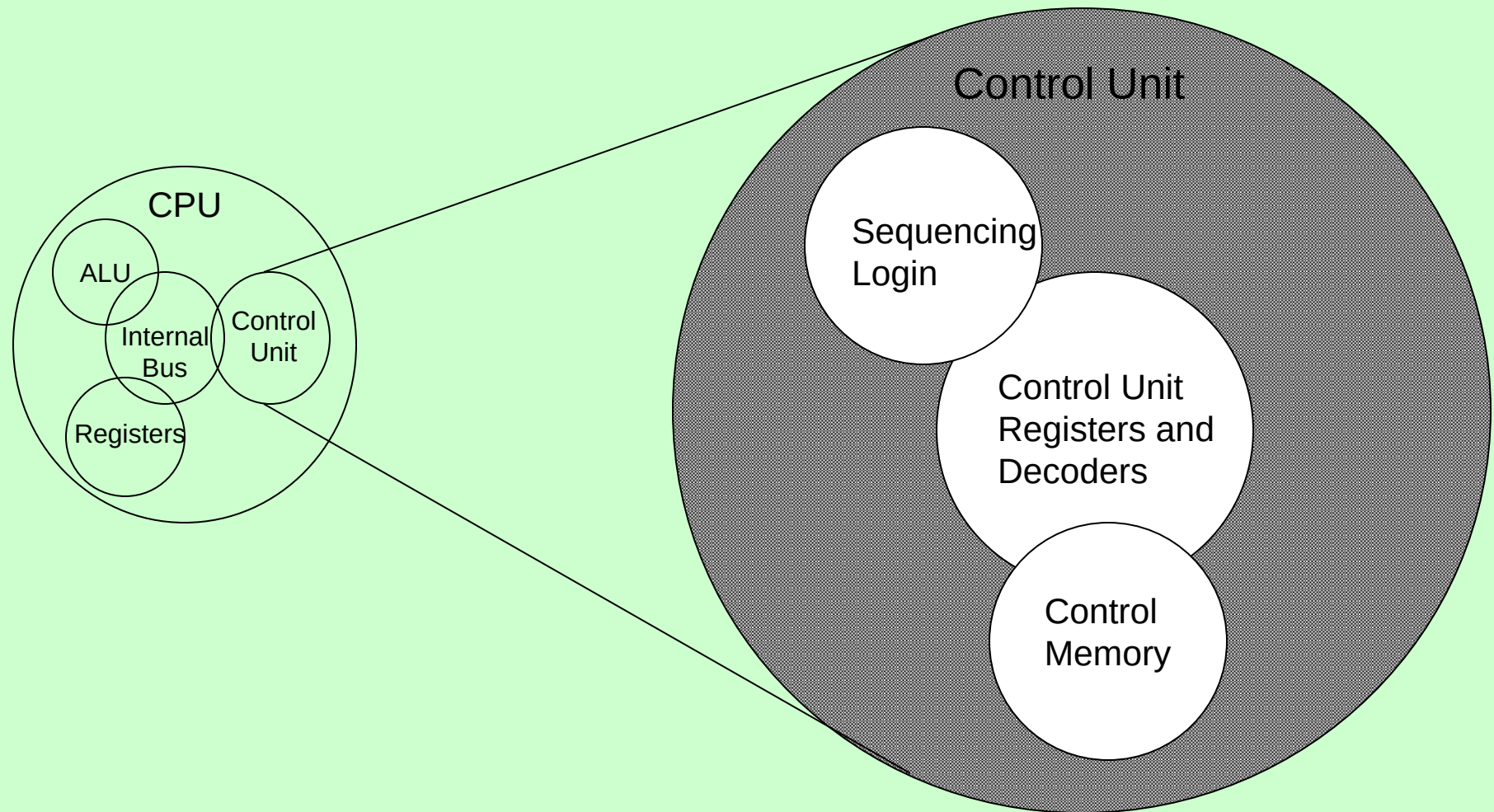
Is the computer component that interprets and executes instructions

Referred to as a **multicore processor** if it contains **multiple cores**

Structure - The CPU



Structure - The Control Unit



Cache Memory

- Multiple layers of memory **between** the **processor** and **main memory**
- Is **smaller** and **faster** than main memory
- Used to speed up memory access by placing in the cache data from main memory that is likely to be used in the near future
- A **greater performance improvement** may be **obtained** by using multiple levels of cache, with level 1 (L1) closest to the core and additional levels (L2, L3, etc.) **progressively farther from the core**

Registers (1 of 2)

- **Memory buffer register (MBR)**
 - Contains a word to be stored in memory or sent to the I/O unit
 - Or is used to receive a word from memory or from the I/O unit
- **Memory address register (MAR)**
 - Specifies the **address** in memory of the word to be written from or read into the MBR
- **Instruction register (IR)**
 - Contains the 8-bit **opcode instruction** being executed

Registers (2 of 2)

- **Instruction buffer register (IBR)**
 - Employed to temporarily hold the right-hand instruction from a word in memory
- **Program counter (PC)**
 - Contains the address of the next instruction pair to be fetched from memory
- **Accumulator (AC) and multiplier quotient (MQ)**
 - Employed to temporarily hold operands and results of ALU operations

Integrated Circuits (1 of 3)

- A computer consists of gates, memory cells, and interconnections among these elements
- The **gates** and **memory** cells are constructed of simple **digital electronic** components

Integrated Circuits (2 of 3)

- **Data storage** – provided by memory cells
- **Data processing** – provided by **gates**
- **Data movement** – the paths among components are used to move data from memory to memory and from memory through gates to memory
- **Control** – the paths among components can carry **control signals**

Integrated Circuits (3 of 3)

- Exploits the fact that such components as **transistors**, **resistors**, and **conductors** can be fabricated from a **semiconductor** such as **silicon**
- Many transistors can be produced at the same time on a **single wafer of silicon**
- Transistors can be connected with a processor metallization to form circuits

Embedded Systems (1 of 2)

- The use of **electronics** and ***software*** within a product
- Billions of computer systems are produced each year that are embedded within larger devices
- Today many devices that use electric power have an embedded computing system



Embedded Systems (2 of 2)

- Often embedded systems are tightly coupled to their **environment**
 - This can give rise to real-time constraints imposed by the need to interact with the environment
 - Constraints such as required speeds of motion, required precision of measurement, and required time durations, dictate the timing of software operations
 - If multiple activities must be managed simultaneously this imposes more complex real-time constraints

The Internet of Things (IoT) (1 of 3)

- Term that refers to the expanding interconnection of smart devices, ranging from appliances to tiny *sensors*
- Is primarily driven by deeply embedded devices
- Generations of deployment culminating in the IoT:
 - **Information technology (IT)**
 - PCs, servers, routers, firewalls, and so on, bought as IT devices by enterprise IT people and primarily using wired connectivity

The Internet of Things (IoT) (2 of 3)

—Operational technology (OT)

- Machines/appliances with embedded IT built by non-IT companies, such as medical machinery, **SCADA**, process control, and kiosks, bought as appliances by enterprise OT people and primarily using wired connectivity

—Personal technology

- **Smartphones, tablets**, and eBook readers bought as IT devices by consumers exclusively using wireless connectivity and often multiple forms of wireless connectivity

The Internet of Things (IoT) (3 of 3)

—Sensor/actuator technology

- Single-purpose devices bought by consumers, IT, and OT people exclusively using wireless connectivity, generally of a single form, as part of larger systems
- It is the fourth generation that is usually thought of as the IoT and it is marked by the use of billions of embedded devices

Embedded Operating Systems

- There are two general approaches to developing an embedded operating system (OS):
 - Take an existing OS and adapt it for the embedded application
 - Design and implement an OS intended solely for embedded use



Application Processors versus Dedicated Processors

- **Application processors**

- Defined by the processor's ability to execute complex operating systems
- General-purpose in nature
- An example is the **smartphone** - the embedded system is designed to support numerous apps and perform a wide variety of functions

- **Dedicated processor**

- Is dedicated to one or a **small number of specific tasks** required by the host device
- Because such an embedded system is dedicated to a **specific task** or tasks, the processor and associated components can be engineered to **reduce size and cost**

Deeply Embedded Systems (1 of 2)

- Subset of embedded systems
- Has a processor whose behavior is difficult to observe both by the **programmer** and the **user**
- Uses a **microcontroller** rather than a **microprocessor**
- Is not programmable once the program logic for the device has been burned into R O M
- Has **no interaction** with a **user**

Deeply Embedded Systems (2 of 2)

- Dedicated, single-purpose devices that detect something in the environment, perform a basic level of processing, and then do something with the results
- Often have **wireless capability** and appear in networked configurations, such as **networks of sensors deployed over a large area**
- Typically have extreme **resource constraints** in terms of **memory, processor size, time, and power consumption**

Review Questions

- **1.** What, in general terms, is the distinction between computer organization and computer architecture?
- **2.** What, in general terms, is the distinction between computer structure and computer function?
- **3.** What are the four main functions of a computer?
- **4.** List and briefly define the main structural components of a computer.
- **5.** List and briefly define the main structural components of a processor.